

Structural Causal Bandits with Non-manipulable Variables

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Abstract

Causal knowledge is sought after throughout data-driven fields due to its explanatory power and potential value to inform decision-making. If the targeted system is well-understood in terms of its causal components, one is able to design more precise and surgical interventions so as to bring certain desired outcomes about. The idea of leveraging the causal understanding of a system to improve decision-making has been studied in the literature under the rubric of *structural causal bandits* (Lee and Bareinboim, 2018). In this setting, (1) pulling an arm corresponds to performing a causal intervention on a set of variables, while (2) the associated rewards are governed by the underlying causal mechanisms. One key assumption of this work is that any observed variable (X) other than the reward is manipulable, which means that intervening and making $do(X = x)$ is always realizable. In many real-world scenarios, however, this requirement can be too stringent. For instance, a decision-maker may be able to change diet but have no intervention that directly sets obesity to a chosen value (Pearl 2018). In this paper, we study a relaxed version of the structural causal bandit problem when not all variables are manipulable. Specifically, we develop a procedure that takes as argument partially specified causal knowledge and identifies the possibly-optimal arms in structural bandits with non-manipulable variables. We further introduce an algorithm that uncovers non-trivial dependence structure among the possibly-optimal arms. Finally, we corroborate our findings with simulations, which show that MAB solvers enhanced with causal knowledge and leveraging the newly discovered dependence structure among arms outperform the corresponding unstructured solvers in the reported settings.

Introduction

Causal knowledge is deemed highly valuable and desirable across a wide range of disciplines, including in the sciences, engineering, and businesses. The reasons for such a prominence are various. First, causal knowledge entails explanatory power, which epitomizes the very goal of science — i.e., opening Nature’s “black box” and explaining how reality “works,” or how it could be understood in terms of more elementary, and interpretable components. Second, and more pragmatically, purposeful agents (AI systems,

policy-makers, physicians) are better equipped for designing cleaner, more precise, and more effective interventions once the underlying causal mechanisms are well-understood (Bareinboim and Pearl 2016; Pearl and Mackenzie 2018; Lee and Bareinboim 2018).

For instance, economists strive to understand the *root causes* of poverty, which could allow the design of new policies (i.e., causal interventions) to improve the population’s socioeconomic status (SES). A considerable body of evidence was accumulated for many decades, notably by the University of Chicago’s Professor, and Nobel Prize laureate, James Heckman, whose work examines the benefits of investment in disadvantaged children (Heckman 2006). The understanding following this causal link translated to the larger support of early childhood education, and a push for new policies; see, for an example, the more than one billion dollars in combined federal, private, and philanthropic commitments announced at the 2014 White House summit (The White House, Office of the Press Secretary 2014). There are many such cases in public health as well — e.g., evidence supports that tobacco smoking is one of the determinant factors of lung cancer (Cornfield 1951; U.S. Department of Health and Human Services 2014), while observational studies report associations between obesity and mortality (Flegal et al. 2005; Pearl 2018).

Despite all the impressive results achieved so far, the treatment of how to use causal knowledge to support decision-making is still in its infancy. If our goal is to build a more automated, data-driven society, and intelligent systems that can reason and act autonomously, we need to move from a heuristic understanding of the interplay between causal knowledge and decision-making (which currently relies on the insights of highly skilled scientists) to a more fundamental understanding of the principles that allow one to translate causal evidence into a more robust decision-making process.

In order to realize this goal, we start by highlighting the rich literature on automated decision-making in AI, including the setting known as *multi-armed bandits* (MABs, for short) (Robbins 1952; Lai and Robbins 1985). We’ll use the MAB framework as the baseline of our analysis. In a typical MAB instance, there are multiple arms (or actions) to play and, at each time step, an arm is pulled and a stochastic reward from the underlying, unknown distribution is returned. Algorithms for the bandit problem attempt to minimize the

cumulative regret, while having to cope with an exploration and exploitation trade-off. This setting has been popular in experimental circles since many real-world situations can be cast as MAB instances, including clinical trials, online advertisement placement, network routing, news article recommendation, just to cite a few. Remarkably, many algorithms with attractive properties have been developed, including the celebrated *Upper Confidence Bound* (Garivier and Cappé 2011; Auer, Cesa-Bianchi, and Fischer 2002; Cappé et al. 2013) and *Thompson sampling* (Thompson 1933; Chapelle and Li 2011; Kaufmann, Korda, and Munos 2012). Standard unstructured solvers do not exploit known relations between arms’ reward distributions. Structured-bandit methods exploit such relations to improve efficiency (Dani, Hayes, and Kakade 2008; Magureanu, Combes, and Proutiere 2014).

Even though not explicitly acknowledged until recently, MAB solvers inherently estimate causal effects, or properties of these effects. To witness, note that (1) pulling an arm corresponds to intervening on a set of variables and setting them to specific values (from whatever *natural* regime the decision used to be based on, to a new compulsory policy determined by the solver); and, (2) the rewards associated with these arms are governed by the underlying causal mechanisms (Bottou et al. 2013; Bareinboim, Forney, and Pearl 2015; Lattimore, Lattimore, and Reid 2016; Zhang and Bareinboim 2017; Forney, Pearl, and Bareinboim 2017; Sen et al. 2017).

Recently, Lee and Bareinboim (2018) combined these observations (i.e., that MABs are inherently causal and that the dependence across arms carries decision-making value) and introduced what they called the *structural causal bandits* problem (SCM-MAB), where a Structural Causal Model (SCM) (Pearl 2000) is used to encode the structural relations among the covariates, arms, outcome variables. Given partial knowledge about the underlying SCM (e.g., the causal graph), the authors characterized two structural properties computable from any SCM-MAB, i.e., i) arms with the same reward distribution using *do*-calculus constraints, and ii) under what topological conditions one arm can be optimal.

For concreteness, we consider the SCM-MAB instance shown in Fig. 1, where Y represents the reward, X and Z represent two variables that can be manipulated, and the dashed-bidirected arrow, following standard notation, represents the existence of a latent common cause (exogenous) affecting both X and Y . There are four sets of variables that can be intervened upon in this case (intervention sets), i.e., $\{\emptyset, \{X\}, \{Z\}, \{X, Z\}\}$, which totals 9 arms if we assume binary variables. The first property reveals that $P(y|do(z)) = P(y|do(x, z))$, which means that intervening on X and Z simultaneously is regarded as wasteful — one should always prefer intervening on $do(Z)$ over $do(X, Z)$. Intuitively, all the effect of X on Y is mediated by Z in this case, which means X has no effect on Y whenever we intervene on Z . The second property characterizes that $\max_z \mu_z \geq \max_x \mu_x$ where $\mu_x = \mathbb{E}[Y|do(x)]$ holds true in *any* model, which means that $do(Z)$ should be *always* preferred over $do(X)$. This implies that three arms, $do(\emptyset)$, $do(z = 0)$, and $do(z = 1)$, relating to the non-dominated, minimal intervention sets, $\emptyset$ and $\{Z\}$, are contenders to be

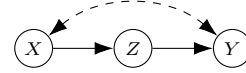


Figure 1: A causal model where Y is the outcome variable.

optimal relative to the underlying SCM compatible with $\mathcal{G}$.¹ Such a minimal set that can be optimal when intervened upon was termed a *possibly-optimal minimal intervention set*, or POMIS, in (Lee and Bareinboim 2018).

In many settings, however, it is not the case that every observed variable is manipulable. For the graph discussed earlier, imagine that X , Z , and Y correspond to diet, cholesterol level, and heart failure. In this illustrative model, suppose the available interventions change diet but cannot directly set cholesterol to a specified value. This is a restriction on the decision-maker’s actions. Given that Z is not manipulable, then intervening on X , e.g., diet, may lead to the desired outcome. Clearly, POMISs under non-manipulable variables will not be, in general, a subset of the POMISs with only manipulable variables. We’ll show in the paper that $do(X)$, which was dominated by $do(Z)$, should be regarded as a POMIS whenever Z is not manipulable. It is critical, therefore, to be able to identify the POMISs from a causal graph under an arbitrary set of non-manipulable variables.

Previous work considered identifying the POMISs based on the graph structure, which allowed a more precise decision-making by the corresponding MAB solver. Still, the dependence structure across arms was not fully exploited. More sophisticated relations among arms can be learned through the *do*-calculus (Pearl 2000, Ch. 3). For instance, the identity $P(y|do(x)) = \sum_z P(z|x) \sum_{x'} P(y|z, x')P(x')$ holds in Fig. 1, which is known as the ‘front-door’ formula. The front-door implies that the distribution of the arm $do(X = x)$ can be learned from $do(\emptyset)$, i.e., without having to directly intervene on X . More generally, an interventional distribution can be expressed in terms of the other distributions, which implies new learning opportunities we aim to leverage.

Specifically, our contributions are as follows:

1. We start by formalizing the SCM-MAB problem under non-manipulability constraints using the language of structural causality. We then characterize the POMIS with manipulability constraints, i.e., the collection of interventions that are possibly optimal.
2. We develop a new algorithm that derives an expression for the arm’s distribution given an arbitrary set of interventional distributions (other arms). We use this result to enhance standard MAB algorithms with causal capabilities, which will now exploit the structural properties of the underlying causal graph and the relations across interventional distributions, improving their efficiency.

The simulations show lower cumulative regret for the solvers that use the causal structure and the available identifying expressions.

¹Note that this implies that not intervening, and letting the system operate in its natural state, can yield an optimal reward.

Preliminaries

We follow the convention in the field and use a capital letter as a variable, X , and the corresponding lower case, x , as its realization. Bold face is used for a set of variables or values, $\mathbf{X}$ or $\mathbf{x}$, blackboard bold for a set of sets, $\mathbb{X}$. We denote by $\mathfrak{X}_X$ the domain of X . We also use calligraphic letters for graphs and models. The basic semantical framework of our analysis relies on Structural Causal Models (SCM, Pearl, 2000), i.e.,

Definition 1 (SCM). A structural causal model (SCM) $\mathcal{M}$ is a 4-tuple $\langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P(\mathbf{U}) \rangle$ where:

- $\mathbf{U}$ is a set of exogenous (unobserved) variables, which are determined by factors outside of the model;
- $\mathbf{V}$ is a set $\{V_i\}_{i=1}^n$ of endogenous (observed) variables that are determined by variables in $\mathbf{U} \cup \mathbf{V}$;
- $\mathbf{F}$ is a set of structural functions $\{f_i\}_{i=1}^n$ where each f_i is a process by which V_i is assigned a value $v_i \leftarrow f_i(\mathbf{pa}^i, \mathbf{u}^i)$ in response to the current values of $\mathbf{PA}^i \subset \mathbf{V}$ and $\mathbf{U}^i \subseteq \mathbf{U}$;
- $P(\mathbf{U})$ is a distribution over the exogenous variables $\mathbf{U}$.

A SCM induces observational and interventional probability distributions, $P(\mathbf{v}) = \sum_{\mathbf{u}} \prod_i P(v_i | \mathbf{pa}^i, \mathbf{u}^i) P(\mathbf{u})$, and $P(\mathbf{v} | \mathbf{t} | do(\mathbf{t})) = \sum_{\mathbf{u}} \prod_{i|V_i \notin \mathbf{t}} P(v_i | \mathbf{pa}^i, \mathbf{u}^i) P(\mathbf{u})$. For short, we write $P(\mathbf{y} | do(\mathbf{x}))$ as $P_{\mathbf{x}}(\mathbf{y})$.

Each SCM is associated with a causal graph $\mathcal{G} = \langle \mathbf{V}, \mathbf{E} \rangle$, where there are two types of edges in $\mathbf{E}$ — directed edges such as $V_i \rightarrow V_j$ indicating direct functional dependence, i.e., V_i is used to define $f_j \in \mathbf{F}$; bidirected edges such as $V_i \leftrightarrow V_j$ indicating the existence of an unobserved confounder (UC, for short), i.e., there exists $U \in \mathbf{U}$ which appears in both f_i and f_j . We use family relations, *pa*, *ch*, *an*, *de* to denote parents, children, ancestors, and descendants of its argument. With *Pa*, *Ch*, *An*, *De*, we include the arguments, e.g., $An(W) = an(W) \cup \{W\}$. Given a set of variables as argument, we use the union of individual outputs, e.g., $An(\mathbf{W}) = \bigcup_{W \in \mathbf{W}} An(W)$. Note that $pa(V_i) = \mathbf{PA}^i$. We denote by $\mathcal{G}_{\overline{\mathbf{X}}}$ the graph with all edges into $\mathbf{X}$ removed.

We assume acyclic SCMs with independent exogenous sources; all shared sources are represented by bidirected edges. Compatibility allows unused permitted arguments and does not require faithfulness. Graph-only POMIS existence ranges over freely chosen finite state spaces, including those of non-manipulable variables. The projection argument below does not establish completeness for fixed cardinalities. A shared latent may have more than two children; the witness construction chooses pairwise sources without restricting the unknown SCM to that form. The bandit algorithms require finitely many arms and Bernoulli rewards. Identifying expressions involving conditional probabilities are evaluated only under the support conditions that make their denominators positive.

The *do*-calculus (Pearl 1995) consists of three rules relating observational and interventional distributions that obey the topology of the causal graph $\mathcal{G}$. For example, $P_{x,z}(y) = P_z(y)$ (Fig. 1) can be inferred due to $(Y \perp\!\!\!\perp X | Z)_{\mathcal{G}_{\overline{\mathbf{X}, \mathbf{Z}}}}$, which is a *d*-separation statement between Y and X given Z in the subgraph $\mathcal{G}_{\overline{\mathbf{X}, \mathbf{Z}}}$. We refer readers to (Pearl 2000, Sec. 3.4) for a more detailed discussion on SCMs and *do*-calculus.

A stochastic K -armed bandit has a fixed reward distribution for each arm and receives a fresh draw on each pull. An unstructured solver does not exploit known relations between these distributions. In this setting, the task is often minimizing Reg_T , the cumulative regret after T rounds. This is the difference between a maximum expected cumulative reward and an expected cumulative reward of a MAB algorithm, $\text{Reg}_T = T\mu^* - \sum_{t=1}^T \mathbb{E}[Y_{a_t}]$, where a_t is the arm played at time t , Y_{a_t} is a random variable associated with the arm, and μ^* is the optimal expected reward. In a SCM-MAB setting, each arm corresponds to intervening on a set of variables. Given $\mathcal{G}$ and Y , *all* arms are $\{\mathbf{x} \in \mathfrak{X}_{\mathbf{X}} \mid \mathbf{X} \subseteq \mathbf{V} \setminus \{Y\}\}$ with $P(Y_{\mathbf{x}})$, the distribution of a reward variable with arm $\mathbf{x}$, is equivalent to an interventional distribution $P_{\mathbf{x}}(Y)$, and $\mu_{\mathbf{x}} = \mathbb{E}[Y | do(\mathbf{x})]$. We denote by $\mu_{\mathbf{x}^*}$ the best expected reward by intervening on $\mathbf{X}$, i.e., $\mu_{\mathbf{x}^*} = \max_{\mathbf{x} \in \mathfrak{X}_{\mathbf{X}}} \mu_{\mathbf{x}}$.

SCM-MAB with Non-manipulability

In this section, we generalize SCM-MABs by allowing non-manipulability constraints, and then characterize the POMIS under such constraints. We start by denoting a set of non-manipulable variables by $\mathbf{N} \subseteq \mathbf{V} \setminus \{Y\}$, where the reward variable Y is assumed to be non-manipulable. For simplicity, we override the definitions of MIS and POMIS by incorporating $\mathbf{N}$ into them, and limiting the intervention set $\mathbf{X}$ to be $\mathbf{X} \subseteq \mathbf{V} \setminus \{Y\} \setminus \mathbf{N}$ instead of $\mathbf{X} \subseteq \mathbf{V} \setminus \{Y\}$. First, we define Minimal Intervention Sets, which represent a non-redundant set of intervention sets.

Definition 2 (Minimal Intervention Set (MIS)). Given $\langle \mathcal{G}, Y, \mathbf{N} \rangle$, a set of variables $\mathbf{X} \subseteq \mathbf{V} \setminus \{Y\} \setminus \mathbf{N}$ is said to be a *minimal intervention set* if there is no $\mathbf{X}' \subset \mathbf{X}$ such that $\mu_{\mathbf{x}'} = \mu_{\mathbf{x}}$ for every assignment $\mathbf{x} \in \mathfrak{X}_{\mathbf{X}}$ and every SCM conforming to $\mathcal{G}$, where $\mathbf{x}'$ is the restriction of $\mathbf{x}$ to $\mathbf{X}'$.

We denote by $\mathbb{M}_{\mathcal{G}, Y}^{\mathbf{N}}$ a set of MISs given $\langle \mathcal{G}, Y, \mathbf{N} \rangle$ where we omit $\mathbf{N}$ if $\mathbf{N} = \emptyset$. The following relationship can be easily derived from the definition, $\mathbb{M}_{\mathcal{G}, Y}^{\mathbf{N}} = \{\mathbf{W} \in \mathbb{M}_{\mathcal{G}, Y} \mid \mathbf{W} \cap \mathbf{N} = \emptyset\}$. MISs can be identified by checking whether any two probability distributions are equal (through Rule 3 of *do*-calculus). Next, we define a subset of the MISs that can lead to an optimal reward.

Definition 3 (Possibly-Optimal Minimal Intervention Set (POMIS)). Given $\langle \mathcal{G}, Y, \mathbf{N} \rangle$, let $\mathbf{X} \in \mathbb{M}_{\mathcal{G}, Y}^{\mathbf{N}}$. If there exists a SCM conforming to $\mathcal{G}$ such that $\mu_{\mathbf{x}^*} > \mu_{\mathbf{w}^*}$ for every $\mathbf{W} \in \mathbb{M}_{\mathcal{G}, Y}^{\mathbf{N}} \setminus \{\mathbf{X}\}$, then $\mathbf{X}$ is a *possibly-optimal minimal intervention set* with respect to $\langle \mathcal{G}, Y, \mathbf{N} \rangle$.

We similarly denote by $\mathbb{P}_{\mathcal{G}, Y}^{\mathbf{N}}$ a set of POMISs given $\langle \mathcal{G}, Y, \mathbf{N} \rangle$. Lee and Bareinboim (2018) introduced two key concepts that will be instrumental to our analysis, i.e., *minimal unobserved confounders' territory* (MUCT) and *interventional border* (IB). MUCT is a minimal set of variables in $An(Y)$, which includes Y and is closed under descendants and unobserved confounders given a causal graph. For example, the MUCT in Fig. 2a includes all the variables as B is confounded with Y , A is confounded with B , and C is a descendant of B (or A). The IB is defined as the parents of the MUCT excluding the MUCT itself, which is an empty set

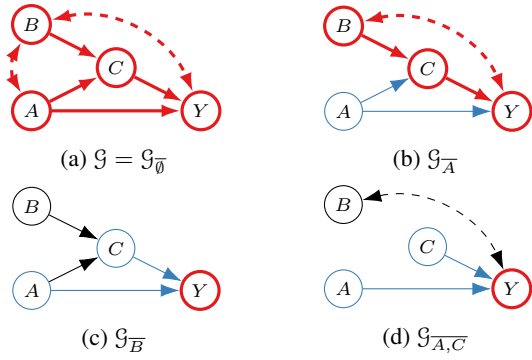


Figure 2: MUCT (thick, red) and IB (blue) on manipulated graphs. Given $\mathcal{G}$ and Y , POMISs are $\emptyset$, $\{A\}$, and $\{A, C\}$.

in this case. With $\mathcal{G}_{\bar{A}}$, the MUCT is $\{B, C, Y\}$ and the IB is $\{A\}$ as shown in Fig. 2b. It was shown that $\mathbf{X} \subseteq \mathbf{V} \setminus \{Y\}$ is a POMIS if and only if IB in $\mathcal{G}_{\bar{\mathbf{X}}}$ is $\mathbf{X}$. For example, $\{A, C\}$ is a POMIS in Fig. 2d while $\{B\}$ is not in Fig. 2c, which indicates that intervening on $\{A, C\}$ is preferred to $\{B\}$.

Characterizing POMIS with Non-manipulability

Identifying all the POMISs given non-manipulable variables is non-trivial as the unconstrained POMIS (i.e., $\mathbf{N} = \emptyset$) cannot be, in general, applicable by retaining only intervention sets containing manipulable variables, i.e., $\exists \mathcal{G}, Y, \mathbf{N} \mathbb{P}_{\mathcal{G}, Y}^{\mathbf{N}} \neq \{\mathbf{X} \in \mathbb{P}_{\mathcal{G}, Y}^{\emptyset} \mid \mathbf{X} \cap \mathbf{N} = \emptyset\}$. This contrasts with MISs in which a set of constrained MISs is a mere subset of unconstrained MISs. Although the equality does not hold in general, the feasible subset of unconstrained POMISs is related to POMIS with constraints as shown below:

Proposition 1. *If $\mathbf{X} \in \mathbb{P}_{\mathcal{G}, Y}$ and $\mathbf{X} \cap \mathbf{N} = \emptyset$, then $\mathbf{X} \in \mathbb{P}_{\mathcal{G}, Y}^{\mathbf{N}}$.*

Simply put, if an arm $do(\mathbf{x})$ is optimal among all arms without constraints, then the arm is still optimal among the manipulable subset of arms. However, this proposition only answers the intersection between $\mathbb{P}_{\mathcal{G}, Y}$ and $\mathbb{P}_{\mathcal{G}, Y}^{\mathbf{N}}$. At this point, the natural question is how to systematically find the POMISs that are not unconstrained POMISs. Recall that in the front-door graph, we noted that $\{X\}$ is a POMIS when $\{Z\}$ is non-manipulable, while this isn't the case in the unconstrained setting. To explain this phenomenon, we note that X , which affects Y through Z , can be seen as directly connected to Y in the constrained setting; clearly, Z should be disregarded since it no longer can disturb the pathway from X to Y .

Focusing only on a subset of variables in a causal graph and their causal relationship lies at the heart of causal modeling. Latent projection (projection, for short) (Verma and Pearl 1990; Verma 1992) is an operation that induces a causal graph over a subset of the variables from the original graph while maintaining some key topological properties of the original causal structure (Tian 2002). In general, the causal graph at hand is already the result of the projection of some unknown, more involved causal structure. For example, the front-door graph (Fig. 1) with the cholesterol example might be the projection of a larger graph such as in Fig. 3, where

the shaded nodes were marginalized out. At an intuitive level, without constraints, POMISs in the front-door graph can be understood as POMISs in the larger graph where the shaded nodes are not manipulable. It would be natural to ask whether we can obtain $\mathbb{P}_{\mathcal{G}, Y}^{\mathbf{N}}$ indirectly from $\mathbb{P}_{\mathcal{H}, Y}$, where $\mathcal{H}$ is the projection of $\mathcal{G}$ onto $\mathbf{V} \setminus \mathbf{N}$.

We formally describe how a projection of a causal graph $\mathcal{G}$ onto $\mathbf{V} \setminus \mathbf{N}$ is constructed next. Let $\hat{\mathcal{G}}$ be a DAG that explicitly represents all exogenous sources, including private noise variables. We initialize a graph $\mathcal{H} = \langle \mathbf{V} \setminus \mathbf{N}, \emptyset \rangle$, then proceed to add

1. a directed edge between V_i and V_j if $V_i \rightarrow V_j \in \mathcal{G}$ or there exists a directed path from V_i to V_j where all non-end nodes in the path between them are in $\mathbf{N}$.
2. a bidirected edge between V_i and V_j if $V_i \leftrightarrow V_j \in \mathcal{G}$; or there is a path in the expanded DAG $\hat{\mathcal{G}}$ with arrowheads at V_i and V_j whose internal vertices are all noncolliders outside $\mathbf{V} \setminus \mathbf{N}$. This includes common causes in $\mathbf{N}$ and exogenous sources that originally affected only one observed variable.

Let $\mathcal{G}_{[\mathbf{V}']}$ be a causal graph obtained by projecting $\mathcal{G}$ onto $\mathbf{V}'$. We will show $\mathbb{P}_{\mathcal{G}, Y}^{\mathbf{N}} = \mathbb{P}_{\mathcal{H}, Y}$ using forward substitution and the realization of a strict-optimality witness for each projected POMIS. Equality of the full SCM classes is not required.

Proposition 2. *Given a SCM $\mathcal{M}^1 = \mathcal{M} = \langle \mathbf{U}, \mathbf{V}, \mathbf{F}, P(\mathbf{U}) \rangle$, there exists a SCM $\mathcal{M}^2 = \mathcal{M}_{[\mathbf{V} \setminus \mathbf{N}]} = \langle \mathbf{U}, \mathbf{V} \setminus \mathbf{N}, \mathbf{F}', P(\mathbf{U}) \rangle$ such that $P_{\mathbf{x}}^1(\mathbf{y}) = P_{\mathbf{x}}^2(\mathbf{y})$ for any $\mathbf{X}, \mathbf{Y} \subseteq \mathbf{V} \setminus \mathbf{N}$ and $\mathbf{Y} \neq \emptyset$, with $\mathbf{X} \cap \mathbf{Y} = \emptyset$.*

Proof. Let $\mathcal{M}' = \mathcal{M}_{[\mathbf{V} \setminus \{W\}]}$ where $W \in \mathbf{N}$. The functions $\{f_X\}_{X \notin ch(W)_{\mathcal{G}}}$ can be used without modifications in $\mathcal{M}'$. We modify functions of the children of W . For every $Q \in ch(W)_{\mathcal{G}}$, we devise f'_Q with f_Q and f_W . By the projection's construction, $\mathbf{PA}'^Q = (\mathbf{PA}^Q \setminus \{W\}) \cup \mathbf{PA}^W$ and let $\mathbf{U}'^Q = \mathbf{U}^Q \cup \mathbf{U}^W$ so that f'_Q takes $\mathbf{pa}^W$ and $\mathbf{u}^W$ as argument in addition to $\mathbf{pa}^Q \setminus \{W\}$ and $\mathbf{u}^Q$, which conforms to $\mathcal{G}_{[\mathbf{V} \setminus \{W\}]}$. Hence, f'_Q is simply f_Q with w computed inside,

$$f'_Q(\mathbf{pa}'^Q, \mathbf{u}'^Q) = f_Q(\mathbf{pa}^Q \setminus \{W\}, f_W(\mathbf{pa}^W, \mathbf{u}^W), \mathbf{u}^Q).$$

This guarantees that $\mathcal{M}_{[\mathbf{V} \setminus \{W\}]}$ and $\mathcal{M}$ yield the same observational and interventional distributions over $\mathbf{V} \setminus \{W\}$. Iterating over the remaining variables in $\mathbf{N}$ proves preservation of the retained intervention distributions. A source may acquire more than two retained children, so this statement does not imply a representation by independent pairwise sources. $\square$

Fig. 4a is an example where marginalizing out A from Fig. 2a induces bidirected edges between B and C (due to the UC between A and B and $A \rightarrow C$), between C and Y since $C \leftarrow A \rightarrow Y$.² Based on Prop. 2, C and Y take the UC between A and B to compute a (the value A would take inside). This explains all three bidirected edges in Fig. 4a.

²Although $B \leftrightarrow A \rightarrow Y$ implies a bidirected edge between B and Y , we do not represent multiple edges of the same type.

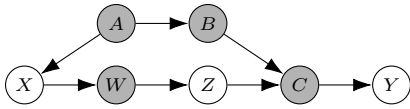


Figure 3: A causal graph where its projection onto $\{X, Y, Z\}$ yields the front-door graph (Fig. 1).

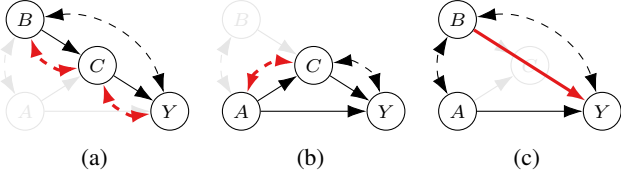


Figure 4: Causal graphs after marginalizing out A , B , and C from Fig. 2a, respectively, with added edges in thick, red.

For a model $\mathcal{M}$, write $J_{\mathcal{M}}(\mathbf{X}) = \max_{\mathbf{x}} \mathbb{E}_{\mathcal{M}}[Y \mid do(\mathbf{x})]$.

Proposition 3 (Preservation of POMIS witnesses). *Let $\mathbf{R} = \mathbf{V} \setminus \mathbf{N}$, with $Y \in \mathbf{R}$, and $\mathcal{H} = \mathcal{G}_{[\mathbf{R}]}$. Under graph compatibility with freely chosen finite state spaces, each $\mathbf{X} \in \mathbb{P}_{\mathcal{H}, Y}$ has a witnessing SCM $\mathcal{M}_{\mathbf{X}}$ compatible with $\mathcal{G}$ such that*

$$J_{\mathcal{M}_{\mathbf{X}}}(\mathbf{X}) > J_{\mathcal{M}_{\mathbf{X}}}(\mathbf{Z}) \quad \text{for all } \mathbf{Z} \in \mathbb{M}_{\mathcal{G}, Y}^{\mathbf{N}} \setminus \{\mathbf{X}\}.$$

The same $\mathcal{M}_{\mathbf{X}}$ serves all competing sets and intervention values.

Proof. A MIS is characterized by a directed path from each of its vertices to Y avoiding the other intervention vertices. Projection contracts paths through $\mathbf{N}$, so the feasible MISs of $\mathcal{G}$ coincide with the MISs of $\mathcal{H}$.

We first choose a particular witness $\mathcal{K}_{\mathbf{X}}$ on $\mathcal{H}$. The border characterization and repaired constructive proofs in (Lee and Bareinboim 2018, Props. 4–5, Thm. 6, Appendix B) supply a finite witness using independent fair bits, each shared by the endpoints of one bidirected edge, together with private fair bits. It satisfies $J_{\mathcal{K}_{\mathbf{X}}}(\mathbf{X}) = 1$ and $J_{\mathcal{K}_{\mathbf{X}}}(\mathbf{Z}) \leq 1/2$ for every other MIS. This selection is valid even when the model class allows a source with more than two children: the necessity argument uses graph separations valid for such sources, and each resulting border has the stated pairwise witness. Pairwise sources are a property of the chosen witness, not a restriction on every compatible SCM.

Fix an elimination sequence from $\mathcal{G}$ to $\mathcal{H}$ and restore vertices in reverse order. Consider one step restoring W ; parents, children and existing edges refer to the graph immediately before that elimination. Retain every private source and every source for an existing bidirected edge. A new bidirected edge between two children of W can be realized by assigning its source privately to W . For a new edge represented by $A \leftrightarrow W \rightarrow B$, assign its source to A and W . These cases exhaust the new bidirected edges; if there is a choice, fix it once. Preserve the source values and their joint distribution, rather than replacing a shared source by independent copies.

Let L_W index the sources assigned to W . Give W the

finite tuple domain

$$\mathfrak{X}_W = \prod_{A \in pa(W)} \mathfrak{X}_A \times \prod_{\ell \in L_W} \mathfrak{X}_{U_{\ell}},$$

$$f_W(\mathbf{pa}_W, \mathbf{u}_{L_W}) = (\mathbf{pa}_W, \mathbf{u}_{L_W}).$$

A child of W reads from this tuple each projected parent argument introduced by a path $A \rightarrow W \rightarrow Q$, and each source value reassigned to W . It then evaluates its previous structural function on these decoded inputs and its remaining direct inputs. Other permitted inputs can be ignored. Nodes that are not children of W keep their functions. Each source is still independent of the others and occurs directly in at most two functions, allowing the next restoration step.

The construction is fixed before choosing an intervention. For any fixed exogenous vector $\mathbf{u}$ and any intervention on retained variables, intervened nodes have the same assigned values in both models. A topological induction shows that each natural retained node receives exactly the parent and source values used by its previous function. Hence, after all restoration steps,

$$V_{do(\mathbf{a})}^{\mathcal{M}_{\mathbf{X}}}(\mathbf{u}) = V_{do(\mathbf{a})}^{\mathcal{K}_{\mathbf{X}}}(\mathbf{u}) \quad (V \in \mathbf{R})$$

for every allowed intervention $do(\mathbf{a})$ and every $\mathbf{u}$, using the same model and source probability measure throughout. The domains of $\mathbf{R}$ are unchanged. All feasible intervention means, and thus all the required strict comparisons, are preserved. $\square$

Scope of the construction. Only the chosen witnesses need to be realized on $\mathcal{G}$. An ordinary bidirected triangle can admit either a shared latent parent or independent pairwise sources, whose distributional model classes need not coincide (Evans 2016); arbitrary SCM restoration is not required here. The tuple construction uses freely chosen domains for $\mathbf{N}$ and does not establish the result for prescribed cardinalities such as binary non-manipulable variables.

We now revisit Fig. 4a for another example. We focus on how the unobserved variables will be assigned. Let UCs be U_{BC} , U_{CY} , and U_{BY} . Unobserved variables other than UCs will be reused as described above. First, $f_B = f'_B$ which already takes U_{BC} and U_{BY} . f_Y takes U_{BY} since $B \leftrightarrow Y$ in $\mathcal{G}$. Since two red bidirected edges do not appear in $\mathcal{G}$, f_A will take both U_{BC} and U_{CY} , and f_C and f_Y receive the relevant source values through the tuple stored at A . Then, we can observe that A is confounded with B via U_{BC} , and B and Y are confounded with U_{BY} . However, U_{CY} will be served as a variable-specific hidden variable for A , which we do not explicitly represent. The construction conforms to $\mathcal{G}$.

Theorem 4. *Given a causal diagram $\mathcal{G} = \langle \mathbf{V}, \mathbf{E} \rangle$, a reward variable $Y \in \mathbf{V}$, and a set of non-manipulable variables $\mathbf{N} \subseteq \mathbf{V} \setminus \{Y\}$, let $\mathcal{H}$ be the projection of $\mathcal{G}$ onto $\mathbf{V} \setminus \mathbf{N}$. For the graph-only POMIS definition with freely chosen finite state spaces,*

$$\mathbb{P}_{\mathcal{G}, Y}^{\mathbf{N}} = \mathbb{P}_{\mathcal{H}, Y}$$

Proof. Projection preserves directed paths between retained vertices, including after interventions on retained sets. Thus

the feasible MISs of $\mathcal{G}$ coincide with the MISs of $\mathcal{H}$. If such a MIS $\mathbf{X}$ is not a graphical POMIS in $\mathcal{H}$, the necessity proof of (Lee and Bareinboim 2018, Thm. 6 and Appendix B) gives the distinct border $\text{IB}(\mathcal{H}_{\overline{\mathbf{X}}}, Y)$ whose best intervention weakly dominates $\mathbf{X}$ in every compatible projected model. To see the dominance directly, put $\mathbf{B} = \text{IB}(\mathcal{H}_{\overline{\mathbf{X}}}, Y)$ and $\mathbf{B}' = \mathbf{B} \setminus \mathbf{X}$. The territory's closure under descendants and bidirected paths implies $(Y \perp\!\!\!\perp \mathbf{B}' \mid \mathbf{X})_{\mathcal{H}_{\overline{\mathbf{X}}\mathbf{B}'}}$. After intervening on the border, no member of $\mathbf{X} \setminus \mathbf{B}$ has a directed path to Y . Rules 2 and 3 yield

$$\mu_{\mathbf{x}} = \sum_{\mathbf{b}'} \mathbb{E}[Y \mid \text{do}(\mathbf{x}[\mathbf{B}], \mathbf{b}')] P_{\mathbf{x}}(\mathbf{b}') \leq \max_{\mathbf{b}} \mu_{\mathbf{b}}.$$

Here $\mathbf{x}[\mathbf{B}]$ is the restriction of $\mathbf{x}$ to $\mathbf{X} \cap \mathbf{B}$. These separations hold in the expanded DAG even if a source has more than two children. The border is a MIS since each of its vertices has a path to Y through the territory avoiding every other border vertex. By Prop. 2, the comparison holds in every original model, so $\mathbf{X}$ cannot be a constrained POMIS. Conversely, Prop. 3 supplies, for each POMIS of $\mathcal{H}$, one compatible SCM on $\mathcal{G}$ in which that set strictly outperforms every other feasible MIS. It is therefore a constrained POMIS of $\mathcal{G}$. This proves the equality without assuming equality of the full distributional model classes. $\square$

In words, the projection onto the manipulable variables together with the reward preserves the causal structure among them with respect to POMIS. Fig. 4 illustrates three projected graphs of $\mathcal{G}$ in Fig. 2a with $\{A\}$, $\{B\}$, and $\{C\}$ marginalized out, respectively. Applying the (unconstrained) POMIS identification procedure (Lee and Bareinboim 2018) on the graphs yields: $\mathbb{P}_{\mathcal{G}, Y}^{\{A\}} = \{\emptyset, \{B\}, \{C\}\}$, $\mathbb{P}_{\mathcal{G}, Y}^{\{B\}} = \{\emptyset, \{A\}, \{A, C\}\}$, and $\mathbb{P}_{\mathcal{G}, Y}^{\{C\}} = \{\emptyset, \{A\}, \{A, B\}\}$.

This section studied the problem of which intervention sets can be optimal. Once the POMIS under constraints are formally characterized, the arm set can be reduced while preserving the optimal reward value; the resulting change in learning performance depends on the estimator and exploration strategy.

Interpretation of arm pruning. POMISs preserve the best attainable expected reward. This does not establish an optimal exploration strategy: an excluded intervention can still provide information useful for estimating other arms.

Exploiting the POMISs Structure

In this section, we note that arms exhibit a dependence structure that we exploit, which follows from the underlying causal structure. The previous section investigated two types of dependence among arms (equality and partial-orderedness), which helped identify POMIS arms, $\{\mathbf{x} \in \mathcal{X}_{\mathbf{X}} \mid \mathbf{X} \in \mathbb{P}_{\mathcal{G}, Y}^{\mathbf{N}}\}$. In this section, we explore more sophisticated dependence among arms that allows one arm's reward distribution to be inferred from pulling another arm. Following the convention, we note that when an arm $\mathbf{x}$ is pulled, a full realization of the other variables in the system is observed, e.g., $\mathbf{v} = \langle v_1, v_2, \dots, v_n \rangle$, which is sampled from the joint distribution $P_{\mathbf{x}}(\mathbf{v})$ associated with the pulled arm.

Algorithm 1 $z^2\text{ID}$

```

1: function  $z^2\text{ID}(\mathcal{G}, \mathbf{y}, \mathbf{x}, \mathbb{Z})$ 
2:   if  $\emptyset \notin \mathbb{Z}$  and  $\mathbf{X} = \emptyset$  then
3:     return  $\sum_{\mathbf{v} \setminus \mathbf{y}} \prod_{\mathbf{s}_i \in \text{CC}(\mathcal{G})} \text{sub-}z^2\text{ID}(\mathcal{G}, \mathbf{s}_i, \mathbf{v} \setminus \mathbf{s}_i, \mathbb{Z})$ 
4:   else return  $\text{sub-}z^2\text{ID}(\mathcal{G}, \mathbf{y}, \mathbf{x}, \mathbb{Z})$ 

5: function  $\text{SUB-}z^2\text{ID}(\mathcal{G}, \mathbf{y}, \mathbf{x}, \mathbb{Z})$ 
6:   for  $\mathbf{Z} \in \mathbb{Z}$  s.t.  $\mathbf{Z} \subseteq \mathbf{X}$  and  $\neg \exists \mathbf{Z}' \in \mathbb{Z} \mathbf{Z}' \subset \mathbf{Z} \subseteq \mathbf{X}$  do
7:     return  $\text{ID}(\mathcal{G} \setminus \mathbf{Z}, \mathbf{y}, \mathbf{x} \setminus \mathbf{Z}, P_{\mathbf{z}})$  if not FAIL
8:   if  $\mathbf{V} \setminus \text{An}(\mathbf{Y})_{\mathcal{G}} \neq \emptyset$  then
9:     return  $\text{sub-}z^2\text{ID}(\mathcal{G}[\text{An}(\mathbf{Y})_{\mathcal{G}}], \mathbf{y}, \mathbf{x} \cap \text{An}(\mathbf{Y})_{\mathcal{G}}, \mathbb{Z})$ 
10:  if  $(\mathbf{W} \leftarrow (\mathbf{V} \setminus \mathbf{X}) \setminus \text{An}(\mathbf{Y})_{\mathcal{G}}) \neq \emptyset$  then
11:    return  $\text{sub-}z^2\text{ID}(\mathcal{G}, \mathbf{y}, \mathbf{x} \cup \mathbf{w}, \mathbb{Z})$ 
12:  if  $|\text{CC}(\mathcal{G} \setminus \mathbf{X})| > 1$  then
13:    return  $\sum_{\mathbf{v} \setminus (\mathbf{x} \cup \mathbf{y})} \prod_{\mathbf{s}_i \in \text{CC}(\mathcal{G} \setminus \mathbf{X})} \text{sub-}z^2\text{ID}(\mathcal{G}, \mathbf{s}_i, \mathbf{v} \setminus \mathbf{s}_i, \mathbb{Z})$ 
14:  return FAIL

```

Recall the causal model in Fig. 1 when Z is non-manipulable. There are three POMIS arms $\text{do}(\emptyset)$, $\text{do}(x=0)$, and $\text{do}(x=1)$, where the two $\text{do}(x)$ arms' reward distributions can be expressed as $P_x(y) = \sum_{\mathbf{z}} P(\mathbf{z} \mid x) \sum_{\mathbf{x}'} P(y \mid \mathbf{z}, \mathbf{x}') P(\mathbf{x}')$, which, as noted, is called the front-door formula. Given such an expression, $P_x(y)$ can be estimated not only with a trivial expression $P_x(y) = \sum_{\mathbf{v} \setminus \{y\}} P_x(\mathbf{v})$, from the experimental samples obtained by pulling the arm $\text{do}(x)$, but also with the front-door formula with observational samples from $P(\mathbf{v})$, in this case, by playing the do-nothing arm. This possibility of leveraging samples from other arms is not an exclusive phenomenon to the model in Fig. 1, but can be cast in much broader terms. Given a general graph $\mathcal{G}$, under what conditions can a probability term (a reward distribution, in particular) be turned into an expression with terms of other distributions?

In the causal inference literature, the problem of *identifiability* (Pearl 2000; Tian and Pearl 2002) asks whether the experimental distribution of interest can be uniquely estimated from the observational distribution (do-nothing intervention). Given a causal graph and $P(\mathbf{V})$, there exists a complete algorithm which outputs an expression, if exists, for a quantity of interest, $P_{\mathbf{x}}(\mathbf{y})$ in this case. A more general problem, called *z-identifiability* (Bareinboim and Pearl 2012), asks whether one can identify $P_{\mathbf{x}}(\mathbf{y})$ when a set of experiments (i.e., distributions) are available, $\{P_{\mathbf{Z}'} \mid \mathbf{Z}' \subseteq \mathbf{Z}\}$, for some $\mathbf{Z} \subseteq \mathbf{V}$. That is, there exists a set of manipulable variables $\mathbf{Z}$, and experiments on every subset of manipulable variables are available.

In the SCM-MAB setting with non-manipulability constraints, it is natural to take advantage of z-identifiability where $\mathbf{Z} = \mathbf{V} \setminus \{Y\} \setminus \mathbf{N}$. Although it is feasible, in a sense, that the resulting expression will not include any of $\mathbf{N}$, the expression may include terms related to experiments on non-POMISs. Consider Fig. 4a for an example: an expression containing $P_{b,c}$ will not be useful since arms $\{\text{do}(b, c)\}_{b \in \mathcal{X}_B, c \in \mathcal{X}_C}$ will not be played at all since $\{B, C\} \notin \mathbb{P}_{\mathcal{G}, Y}^{\mathbf{N}}$. Hence, there exists a need to extend the treatment given to z-identifiability to account for an arbitrary set of experiments, $\{P_{\mathbf{Z}}\}_{\mathbf{Z} \in \mathbb{Z}}$, where $\mathbb{Z}$ is a subset of the

power set of $\mathbf{V}$. We developed a generalized z -identifiability algorithm (Alg. 1), which we call z^2 ID. Using this algorithm, a returned expression establishes how an arm's reward distribution can be estimated from the other arms' information; failure alone does not establish non-identifiability. Before explaining the algorithm, we introduce a few notations. We denote by $\mathcal{G}[\mathbf{W}]$, a vertex-induced subgraph of $\mathcal{G}$ by $\mathbf{W}$ (not to confuse with $\mathcal{G}_{[\mathbf{V}']}$). Also, $\mathcal{G} \setminus \mathbf{W} = \mathcal{G}[\mathbf{V} \setminus \mathbf{W}]$. $\text{CC}(\mathcal{G})$ represents C -components (Tian and Pearl 2002) of $\mathcal{G}$, which is a family of sets of endogenous variables where each set consists of variables that are maximally connected via bidirected edges.

z^2 ID first examines whether the query of interest is an observational quantity where only interventional distributions are available (Line 2). Then, the procedure tries to identify the quantity by decomposing it into multiple interventional quantities using C -component factorization (Tian and Pearl 2002) (Line 3). This part, especially, differs from variants of identifiability algorithms where an observational distribution $P(\mathbf{V})$ is taken for granted. Then, z^2 ID recursively transforms the query $P_{\mathbf{x}}(y)$ into different forms using well-known equalities that can be derived from basic probability operations and *do*-calculus, and it tries to reduce $P_{\mathbf{x}'}(y)$ to an expression only made of $P_{\mathbf{z}}$ whenever $\mathbf{Z} \subseteq \mathbf{X}'$ and $\mathbf{Z} \in \mathbb{Z}$. This can be understood as solving an identifiability problem where the query is $Q_{\mathbf{x}' \setminus \mathbf{z}}(y) = P_{\mathbf{x}'}(y)$, and $Q(\mathbf{v}) = P_{\mathbf{z}}(\mathbf{v})$ is regarded as an available observational distribution.

We show next that the procedure is indeed sound, i.e., any expression that it returns is a valid equality with respect to the underlying structural causal model.

Theorem 5 (soundness). *Whenever z^2 ID returns an expression for $P_{\mathbf{x}}(\mathbf{y})$, it is correct.*

Proof. This follows from the previous identifiability results (Tian and Pearl 2002; Shpitser and Pearl 2006; Bareinboim and Pearl 2012). Specifically, Lines 2–3 correspond to C -factorization (Tian and Pearl 2002). Lines 7–8 follow from soundness of ID since $P_{\mathbf{x}}(\mathbf{y}) = P_{\mathbf{x} \setminus \mathbf{z}, \mathbf{z}}(\mathbf{y})$ which is identifying $Q_{\mathbf{x} \setminus \mathbf{z}}(\mathbf{y})$ with $Q = P_{\mathbf{z}}$ in $\mathcal{G} \setminus \mathbf{Z}$ (Bareinboim and Pearl 2012). Lines 9–12 are based on *do*-calculus. Finally, Line 13–14 follows from Lemma 3 of (Shpitser and Pearl 2006). $\square$

For concreteness, consider a causal graph $\mathcal{G}$ in Fig. 2a. With $\mathbf{N} = \{A\}$, the POMISs are $\{\emptyset, \{B\}, \{C\}\}$ (as shown in Fig. 4a above). The algorithm z^2 ID returns expressions for each arm's distribution as follows:

$$P(y) = \sum_{a,b,c} P_b(c|a)P_c(a,b,y) \quad (1)$$

$$P_b(y) = \sum_{a,c} P(c|a,b) \sum_{b'} P(y|a,b',c)P(a,b') \quad (2)$$

$$P_c(y) = \sum_{a,b} P(y|a,b,c)P(a,b) \quad (3)$$

$$P_c(y) = \sum_a P_b(y|a,c)P_b(a) \quad (4)$$

Note that there are two equations for $P_c(y)$, Eqs. (3) and (4) — the former suggests that P_c can be estimated with samples from the *do*-nothing intervention, while the second says that it can be estimated from samples from P_b . Moreover, b in Eq. (4) can be any realization within $\mathcal{X}_B$. This means that such a mapping can be viewed as two separate sources of data, i.e., $\sum_a P_{b=0}(y|a,c)P_{b=0}(a)$ and $\sum_a P_{b=1}(y|a,c)P_{b=1}(a)$.

Algorithm 2 bMVWA

```

1: function BMVWA( $\mathbf{D}, \{\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_m\}, B$ )
   Input:  $\mathbf{D}$ : samples  $\{D_{\mathbf{x}}\}_{\mathbf{x} \in \mathbf{A}}; \{\hat{\theta}_1, \hat{\theta}_2, \dots, \hat{\theta}_m\}$ :  $m$  estimators;
    $B \geq 2$ : bootstrap replicates;  $\bar{\theta} = B^{-1} \sum_b \hat{\theta}^{(b)}$ 
   Use evaluable estimators only; if unavailable, return the direct
   beta mean and variance.
2: for  $b \in 1, \dots, B$  do
3:    $\mathbf{D}^{(b)} \leftarrow \{D_{\mathbf{x}}^{(b)}\}_{\mathbf{x} \in \mathbf{A}}$  ▷ bootstrap samples
4:    $(\forall_{i=1}^m) \hat{\theta}_i^{(b)} \leftarrow \hat{\theta}_i(\mathbf{D}^{(b)})$  ▷ evaluate expressions
5:    $\hat{\Sigma} \leftarrow \frac{1}{B-1} \sum_{b=1}^B (\hat{\theta}^{(b)} - \bar{\theta})(\hat{\theta}^{(b)} - \bar{\theta})^T$ 
6:    $\mathbf{w} \leftarrow \arg \min_{\mathbf{w}} \mathbf{w}^T \hat{\Sigma} \mathbf{w}$  such that  $\sum_{i=1}^m w_i = 1, w_i \geq 0$ 
7:   return  $\sum_{i=1}^m w_i \hat{\theta}_i(\mathbf{D}), \mathbf{w}^T \hat{\Sigma} \mathbf{w}$ 

```

The failing condition of z^2 ID implies that it encountered a (decomposed) probability term that cannot be reduced to any of the available distributions *individually*. It is an open problem, and outside of the scope of this work, to prove the necessity of z^2 ID, which requires to show how the available distributions cannot answer the query *collectively*.

Integrating Expressions into MAB Algorithms

Given that we have multiple expressions for each arm's distribution, it is crucial to find a principled way of combining them. Let $\theta = P_{\mathbf{x}}(y)$ be a quantity of interest, and there be m dependent estimators $\hat{\theta} = \{\hat{\theta}_i\}_{i=1}^m$ corresponding to expressions relating to θ . They can be combined using a weighted sum, $\hat{\theta} = \sum_{i=1}^m w_i \hat{\theta}_i$ with $\sum_{i=1}^m w_i = 1$ and $(\forall_{1 \leq i \leq m}) w_i \geq 0$ (i.e., convex combination). For fixed weights, $\text{Var}(\hat{\theta}) = \mathbf{w}^T \Sigma \mathbf{w}$, where Σ is the covariance matrix of the estimators. Minimizing this quantity is a constrained quadratic program. Plug-in estimators of identifying expressions are not generally unbiased, and weights estimated from the same data are random. Thus this calculation motivates minimizing an estimated covariance objective; it does not establish the true minimum variance or unbiasedness of the resulting adaptive estimator. Bootstrap is a well-known technique to estimate the variance of an estimator (Efron and Tibshirani 1993). Given data D of size n , sampling n instances from the data, with replacement, yields a bootstrap sample. By repeating the procedure B times, we obtain bootstrap samples $\{D^{(b)}\}_{b=1}^B$, which allow us to estimate sample variance for an estimator.

Given multiple datasets, we bootstrap them simultaneously, and obtain bootstrap estimates for each estimator so as to compute a covariance matrix Σ . We then compute weights minimizing $\mathbf{w}^T \hat{\Sigma} \mathbf{w}$. The returned value is a bootstrap covariance approximation, conditional on the fitted weights. A pseudo-code (Alg. 2) describes the aforementioned procedure, named bootstrap-based minimum variance weighted average (bMVWA).

We introduce two algorithms z^2 -TS and z^2 -KL-UCB (Alg. 3), generalizations of the classic TS and KL-UCB algorithms, respectively, that incorporate bMVWA so that information about other arms' rewards can be exploited to infer an arm's expected reward more precisely via expressions retrieved from z^2 ID. We first explain z^2 -TS. Given a SCM-

MAB instance with non-manipulable variables ($\mathcal{G}$, Y , and N), the algorithm first computes corresponding POMISs (Line 2) and obtain estimators for arms relevant to the POMISs (Line 4). For each arm $\mathbf{x}$, bMVWA returns a weighted estimate and an approximate variance. Then, it finds a pair of parameters for a beta distribution whose mean and variance correspond to $\hat{\theta}_{\mathbf{x}}$ and $\hat{s}_{\mathbf{x}}^2$, and a sample is drawn from this distribution as a heuristic posterior-like sample; posterior calibration is not established here.³ Similarly to TS, an arm is chosen based on posterior samples.

Next we describe z^2 -KL-UCB. The algorithm follows the same initialization steps, and computes each arm's mean payout and its variance through bMVWA . Conventional KL-UCB computes the upper-confidence bound of the expected reward of an arm $\mathbf{x}$ with a non-decreasing function f and how many times the arm is pulled, $N_{\mathbf{x}}$. It infers $\hat{N}_{\mathbf{x}}$ (Line 18), a variance-based effective sample size for each arm $\mathbf{x}$, rather than its actual number of pulls, $N_{\mathbf{x}}$. This substitution does not inherit the confidence or regret guarantees of standard KL-UCB.

One property of the bootstrap is that if the number of samples is too small, bootstrap will fail, in the sense that the bootstrap distribution will not approximate the target distribution. In practice, the aforementioned approach will be deferred until enough samples have been observed. As a rule of thumb, we do not use an expression (i.e., estimator) unless the denominators needed to evaluate it have adequate empirical support. This finite-sample check cannot establish population positivity. If any shared bootstrap replicate makes a required conditional undefined, the affected estimator is excluded from that call. For example, Eq. (1) will be evaluated if all 8 and 16 values of $P_b(c|a)$ and $P_c(a, b, y)$ have positive empirical counts, respectively (assuming binary variables). This conservative rule is stronger than requiring only the relevant conditioning cells to be nonempty. Furthermore, with a binary reward variable, we utilize a beta distribution, $\text{Beta}(\alpha_{\mathbf{x}} + 1, \beta_{\mathbf{x}} + 1)$, for the mean and variance of a trivial estimator, e.g. $\hat{\theta}_i = P_{\mathbf{x}}(y)$, where $\alpha_{\mathbf{x}}$ and $\beta_{\mathbf{x}}$ are the number of $Y_{\mathbf{x}}$ being 1 and 0, respectively, at round t . The beta mean is a smoothed estimate and is not an unbiased sample mean.

For a mean m and variance v , a beta distribution with positive parameters exists only if $0 < m < 1$ and $0 < v < m(1 - m)$; then $k = m(1 - m)/v - 1$, $\alpha = mk$, and $\beta = (1 - m)k$. If the combined estimates fail these conditions, use the direct arm's $\text{Beta}(\alpha_{\mathbf{x}} + 1, \beta_{\mathbf{x}} + 1)$ mean and variance. This also specifies the empty-data initialization. The same fallback prevents division by zero in the effective sample size. These are implementation safeguards, not coverage guarantees.

Empirical Evaluation

In this section, we evaluate the performance (cumulative regret, CR for short) of SCM-MAB algorithms under different strategies so as to assess the effect of employing POMIS-based arm refinement as well as the dependence structure among arms. In particular, we compare the following strate-

gies: Brute-force (BF), which intervenes on every combination of manipulable variables, MIS $\mathbb{M}_{\mathcal{G}, Y}^N$; POMIS $\mathbb{P}_{\mathcal{G}, Y}^N$; POMIS+, which is POMIS enhanced with z^2 ID. Formally, note that the following relationships hold: $\text{BF} \supseteq \text{MIS} \supseteq \text{POMIS} = \text{POMIS+}$ relative to their arms. Combined with two base MAB algorithms, TS and KL-UCB, we compared a total of eight settings, where only POMIS+ employs z^2 -TS and z^2 -KL-UCB.

For the corrected pseudocode, discard FAIL outputs at initialization, use the support and beta-feasibility rules in the preceding section, and set $f(s) = \max\{0, \ln s + 3 \ln \ln s\}$ for $s > 1$. If $T < |\mathbf{A}|$, truncate initialization after T pulls; if only one arm exists, play it for all T rounds. The original simulation plots and table entries are retained; the experiments have not been rerun with the proposed safeguards.

We simulated three SCM-MAB instances, i.e., the front-door setting (Fig. 1) and two models following Fig. 2a and Fig. 5d. The time horizon T is set to 10,000, which is enough to observe the performance difference; simulations are repeated 1,000 times, and the number of bootstraps B is set to 500. Detailed descriptions and expressions generated by z^2 ID are provided in the full technical report (Lee and Bareinboim 2019). Fig. 5 illustrates experimental results and Table 1 summarizes the average cumulative regrets. The percentages reported below are computed from the displayed, rounded table entries. We discuss these results in the sequel.

(The Front-door Graph with $N=\{Z\}$) The results are shown in Fig. 5a. Due to its simplicity, BF, MIS, and POMIS are all the same in this setting.⁴ A gap between POMIS and POMIS+ indicates how well expressions reduce uncertainty of arms' expected rewards so that the underlying MAB algorithm plays the optimal arm more often (and suboptimal arms less often). POMIS+ reduces CRs 34.5% (TS) and 39.2% (KL-UCB) compared to POMIS.

(Causal Model for Fig. 2a with $N=\{A\}$) Results are shown in Fig. 5b. MIS and POMIS have the same performance as their arms are the same. The gain for (PO)MIS compared to BF is due to the fact that there are four fewer arms corresponding to intervening on $\{B, C\}$. z^2 ID helps reduce cumulative regret for both MAB algorithms. Reductions in CR by employing POMIS+ compared to POMIS are 46.2% and 61.1% for TS and KL-UCB, respectively. Such a high reduction rate is achieved in part because expressions construct reciprocal relationships among arms (see Eqs. (1) to (4)) except that $P_b(y)$ is not expressed with $P_c(v)$.

(Causal Model for Fig. 5d) The results are shown in Fig. 5c. This graph clearly demonstrates the advantage of having a smaller number of arms (BF 27, MIS 19, POMIS 15). In this setting, POMIS+ makes use of 24 expressions. However, the larger number of arms (compared to the previous tasks) and larger number of variables involved in formulas, e.g., a probability term $P(y|a, b, d, e)$ in some formula, make POMIS+ difficult to take full advantage of every expression in a given time horizon $T = 10,000$. The reported time hori-

³One might simply use a weighted average of one of bootstrap estimates, $\mathbf{w}^T \hat{\theta}^{(b)}$ for some b .

⁴If two strategies share the same set of arms, then their results will be exactly the same since we assigned a unique random seed for each of 1,000 Monte Carlo simulations so as to reproduce the same experimental results.

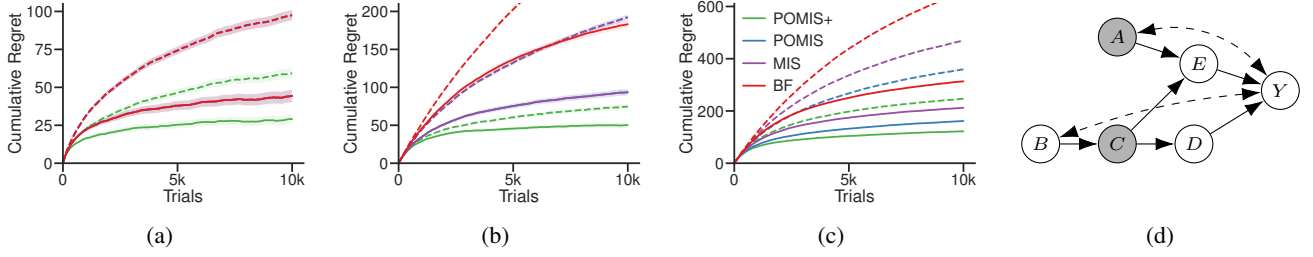


Figure 5: (a,b,c) Simulated results for settings in Figs. 1, 2a, and 5d, where the average cumulative regret (95% confidence interval) is shown relative to the four strategies discussed in the paper (BF, MIS, POMIS, POMIS+). The results are also shown for the corresponding TS (solid) and KL-UCB (dashed) solvers. (d) The causal graph used in (c) with $N = \{A, C\}$.

Algorithm 3 Bernoulli TS and KL-UCB with z^2 ID

```

1: function  $z^2$ -TS( $\mathcal{G}, Y, N, T, B$ )
2:    $\mathbb{Z} \leftarrow \mathbb{P}_{\mathcal{G}, Y}^N$ 
3:    $\mathbf{A} \leftarrow \{\mathbf{x} \in \mathbb{X} \mid \mathbf{X} \in \mathbb{Z}\}$ 
4:    $\hat{\theta}_{\mathbf{x}} \leftarrow \{P_{\mathbf{x}}(y)\} \cup \{z^2\text{ID}(\mathcal{G}, y, \mathbf{x}, \mathbb{Z}')\}_{\mathbb{Z}' \subseteq \mathbb{Z} \setminus \{\mathbf{x}\}}$  for  $\mathbf{x} \in \mathbf{A}$ 
5:    $\mathbf{D} \leftarrow \{D_{\mathbf{x}} = \emptyset\}_{\mathbf{x} \in \mathbf{A}}$ 
6:   for  $t$  in  $1, \dots, T$  do
7:     for  $\mathbf{x} \in \mathbf{A}$  do
8:        $\theta_{\mathbf{x}}, \hat{s}_{\mathbf{x}}^2 \leftarrow \text{bMVWA}(\mathbf{D}, \hat{\theta}_{\mathbf{x}}, B)$ 
9:       Apply the beta-feasibility fallback; find  $\hat{\alpha}_{\mathbf{x}}, \hat{\beta}_{\mathbf{x}}$  such that
         Beta( $\hat{\alpha}_{\mathbf{x}}, \hat{\beta}_{\mathbf{x}}$ ) matching  $\hat{\theta}_{\mathbf{x}}, \hat{s}_{\mathbf{x}}^2$ 
10:       $\theta_{\mathbf{x}} \sim \text{Beta}(\hat{\alpha}_{\mathbf{x}}, \hat{\beta}_{\mathbf{x}})$ 
11:       $\mathbf{x}' \leftarrow \arg \max_{\mathbf{x} \in \mathbf{A}} \theta_{\mathbf{x}}$ 
12:      Sample  $\mathbf{v}$  by  $\text{do}(\mathbf{x}')$  and append  $\mathbf{v}$  to  $D_{\mathbf{x}'}$ 

13: function  $z^2$ -KL-UCB( $\mathcal{G}, Y, N, T, B, f$ )
14:   Initialize  $\mathbb{Z}, \mathbf{A}, \{\hat{\theta}_{\mathbf{x}}\}_{\mathbf{x} \in \mathbf{A}}, \mathbf{D}$ 
15:   ( $\forall \mathbf{x} \in \mathbf{A}$ ) Sample  $\mathbf{v}$  by  $\text{do}(\mathbf{x})$ , and append  $\mathbf{v}$  to  $D_{\mathbf{x}}$ 
16:   for  $t$  in  $|\mathbf{A}| + 1, \dots, T$  do
17:      $\hat{\theta}_{\mathbf{x}}, \hat{s}_{\mathbf{x}}^2 \leftarrow \text{bMVWA}(\mathbf{D}, \hat{\theta}_{\mathbf{x}}, B)$  for  $\mathbf{x} \in \mathbf{A}$ 
18:     Apply the beta-feasibility fallback for every arm
19:      $\hat{N}_{\mathbf{x}} \leftarrow \hat{\theta}_{\mathbf{x}}(1 - \hat{\theta}_{\mathbf{x}})/\hat{s}_{\mathbf{x}}^2$ ;  $\hat{t} \leftarrow \sum_{\mathbf{x}} \hat{N}_{\mathbf{x}}$ 
20:      $\mu = \left\{ \sup \left\{ \mu \in [0, 1] : \text{KL}(\hat{\theta}_{\mathbf{x}}, \mu) \leq \frac{f(\hat{t})}{\hat{N}_{\mathbf{x}}} \right\} \right\}_{\mathbf{x} \in \mathbf{A}}$ 
21:      $\mathbf{x}' \leftarrow \arg \max_{\mathbf{x} \in \mathbf{A}} \mu_{\mathbf{x}}$ 
22:     Sample  $\mathbf{v}$  by  $\text{do}(\mathbf{x}')$ , and append  $\mathbf{v}$  to  $D_{\mathbf{x}'}$ 

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zon does not establish how this difference changes at larger horizons. POMIS+ reduced CRs 24.8% and 31.3% for TS and KL-UCB, respectively, compared to those for POMIS.

In sum, our experiments corroborate our theoretical findings — the tested MAB algorithms benefit from playing a smaller number of qualified arms (POMIS) and precise estimations of arms' rewards (z^2 ID). As per our observations, z^2 ID will be less rewarding if the gap between rewards of the optimal and sub-optimal arms is big enough so that they can easily be distinguished before the MAB algorithm takes advantage of available expressions. Further, the performance gain will not be outstanding if the aforementioned gap is relatively small and the given time horizon is limited, or if there are many arms with low rewards that are not sufficiently played to utilize related expressions.

		Fig. 5a	Fig. 5b	Fig. 5c
TS	POMIS+	29.08	50.29	121.54
	POMIS	44.40	93.53	161.54
	MIS	"	"	212.24
	BF	"	183.15	313.39
UCB	POMIS+	59.30	74.93	247.05
	POMIS	97.52	192.39	359.40
	MIS	"	"	469.30
	BF	"	328.68	657.11

Table 1: Average cumulative regret at $T = 10,000$

Conclusions

We studied the problem of *structural causal bandits* that asks whether (and how) a decision-maker should intervene in a causal system so as to optimize a particular measure. Our results generalize the previous treatment given to the problem (Lee and Bareinboim 2018), which assumed that all observed variables other than the reward are manipulable (i.e., could be intervened upon). For example, a decision-maker may be able to change diet while lacking an action that sets cholesterol directly. The set of feasible interventions must therefore be specified as part of the decision problem. Formally, we characterized, at graph-only scope with unrestricted state spaces, two crucial properties in structural bandits with manipulability constraints, i.e.: (1) the possibly-optimal arms (POMIS), and (2) the topological relationship between such arms. We further developed an identification algorithm that outputs an expression for an (interventional) distribution providing a way to fuse an arbitrary set of experiments so that each arm's reward distribution can be estimated from samples of other arms. We equipped existing MAB algorithms with such capabilities, which are now able to play only a qualified subset of the arms while more accurately estimating their expected rewards. Following the current debate on the topic (Pearl 2018), we hope that our results will help to move the discussion from whether a non-manipulable variable causes another to how a decision-maker in the real world should intervene (or not intervene) in the system and be able to optimize a socially agreed target outcome (e.g., survival, happiness, wealth).

Appendix

We describe three settings and information used in the experimental section. We denote an exclusive-or function by $\oplus$. An expression with $[\dots]_x$ represents that any $x \in \mathfrak{X}_X$ can be assigned.

Frontdoor graph

$\mathbf{N} = \{Z\}$. $\mathcal{M}$ is defined as

- $P(U_X = 1) = 0.5, P(U_Y = 1) = 0.4, P(U_Z = 1) = 0.4, P(U_{XY} = 1) = 0.5$, with $P(\mathbf{U}) = \prod_{U \in \mathbf{U}} P(U)$.
- $f_X = u_X \oplus u_{XY}$
- $f_Z = u_Z \oplus x$
- $f_Y = (u_Y \wedge u_{XY}) \oplus z$

Intervention sets are

- $\text{MIS} = \{\emptyset, \{X\}\}$,
- $\mathbb{P}_{\mathfrak{G}, Y}^{\mathbf{N}} = \{\emptyset, \{X\}\}$

This is a sample of the output of $\mathbf{z}^2\text{ID}$:

$$P_x(y) = \sum_z P(z|x) \sum_{x'} P(y|x', z) P(x')$$

4-variable graph

$\mathbf{N} = \{A\}$. $\mathcal{M}$ is defined as

- $P(U_B = 1) = 0.5, P(U_C = 1) = 0.25, P(U_Y = 1) = 0.25, P(U_{BY} = 1) = 0.25, P(U_{AB} = 1) = 0.4$, with $P(\mathbf{U}) = \prod_{U \in \mathbf{U}} P(U)$.
- $f_A = u_{AB}$
- $f_B = u_B \oplus u_{AB} \oplus u_{BY}$
- $f_C = u_C \oplus a \oplus b$
- $f_Y = 1 - (u_{BY} \oplus u_Y \oplus a \oplus c)$

Intervention sets are

- $\text{MIS} = \{\emptyset, \{B\}, \{C\}\}$,
- $\mathbb{P}_{\mathfrak{G}, Y}^{\mathbf{N}} = \{\emptyset, \{B\}, \{C\}\}$

This is a sample of the output of $\mathbf{z}^2\text{ID}$:

$$P(y) = \sum_{a,b,c} P_b(c|a) P_c(a, b, y)$$

$$P_b(y) = \sum_{a,c} P(c|a, b) \sum_{b'} P(y|a, b', c) P(a, b')$$

$$P_c(y) = \sum_{a,b} P(y|a, b, c) P(a, b)$$

$$P_c(y) = \left[\sum_a P_b(y|a, c) P_b(a) \right]_b$$

6-variable graph

$\mathbf{N} = \{A, C\}$. $\mathcal{M}$ is defined as

- $P(U_A = 1) = 0.22, P(U_B = 1) = 0.2, P(U_C = 1) = 0.12, P(U_D = 1) = 0.2, P(U_E = 1) = 0.04, P(U_Y = 1) = 0.87, P(U_{BY} = 1) = 0.04, P(U_{AY} = 1) = 0.45$, with $P(\mathbf{U}) = \prod_{U \in \mathbf{U}} P(U)$.

- $f_A = u_A \oplus u_{AY}$

- $f_B = u_B \oplus u_{BY}$

- $f_C = u_C \oplus b$

- $f_D = u_D \oplus c$

- $f_E = u_E \oplus a \oplus c$

- $f_Y = 1 - (u_Y \oplus u_{AY} \oplus u_{BY} \oplus d \oplus e)$

Intervention sets are

- $\text{MIS} = \{\emptyset, \{B\}, \{D\}, \{E\}, \{B, D\}, \{B, E\}, \{D, E\}\}$,

- $\mathbb{P}_{\mathfrak{G}, Y}^{\mathbf{N}} = \{\emptyset, \{B\}, \{D\}, \{E\}, \{B, D\}, \{D, E\}\}$

We list below a sample of the identities generated by $\mathbf{z}^2\text{ID}$

with respect to the causal model in Fig. 5d:

$$\begin{aligned}
P(y) &= \sum_{a,b,c,d,e} P_e(y|a,b,d)P_b(e|a,c)P_e(c|b) \\
&\quad \times P_e(d|c)P_e(a,b) \\
P(y) &= \sum_{a,b,c,d,e} P_e(y|a,b,d)P_d(e|a,c)P_e(c|b) \\
&\quad \times P_e(d|c)P_e(a,b) \\
P(y) &= \sum_{a,b,c,d,e} P_{b,d}(e|a,c)P_e(y|a,b,d)P_e(c|b) \\
&\quad \times P_e(d|c)P_e(a,b) \\
P_e(y) &= \sum_{b,c,d} P_{e,d}(c|b)P_{e,d}(b,y)P_b(d|c) \\
P_e(y) &= \sum_{b,d} P(d|b) \\
&\quad \times \sum_a P(y|a,b,d,e)P(a,b) \\
P_b(y) &= \sum_{c,d} P(c,d|b) \sum_{a,e} P(e|a,c) \\
&\quad \times \sum_{b'} P(y|a,b',d,e)P(a,b') \\
P_d(y) &= \sum_{a,b,c,e} P_e(y|a,b,d)P_b(e|a,c)P_e(c|b) \\
&\quad \times P_e(a,b) \\
P_d(y) &= \sum_{a,b,c,e} P_{e,d}(a,b,y)P_{e,d}(c|b)P_b(e|a,c) \\
P_d(y) &= \sum_{b,c} P(c|b) \sum_a P(a,b) \\
&\quad \times \sum_e P(y|a,b,d,e)P(e|a,c) \\
P_{d,e}(y) &= \sum_b P_e(y|b,d)P_e(b) \\
P_{d,e}(y) &= \sum_{a,b} P_d(y|a,b,e)P_d(a,b) \\
P_{d,e}(y) &= \sum_{a,b} P(y|a,b,d,e)P(a,b) \\
P_{b,d}(y) &= \sum_c P(c|b) \sum_{a,e} P(e|a,c) \\
&\quad \times \sum_{b'} P(y|a,b',d,e)P(a,b') \\
P_{b,d}(y) &= \sum_c P_b(c) \sum_a P_b(a) \\
&\quad \times \sum_e P_b(y|a,d,e)P_b(e|a,c)
\end{aligned}$$

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Changes

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These proposed corrections were prepared with AI assistance. They are not a publisher-issued correction; AI assistance does not imply author approval or certify correctness.

Location	Before → After	Assumptions / limits
Proposition 3	Restore every SCM on the projected graph → for each projected POMIS, construct one original-graph SCM witnessing strict superiority to all other feasible MISs.	Unrestricted restoration is false with general shared latent sources. Independent pairwise and private sources belong to the selected witness; the unknown SCM is not restricted to this form.
Preliminaries; Proposition 3	Implicit cardinality requirements → freely chosen finite domains; tuple-valued non-manipulable variables transmit witness inputs and shared sources.	Domains are chosen for the witness; mapping then preserves retained domains and may enlarge non-manipulable domains. Completeness for prescribed, including binary, domains is not established. Unused inputs are allowed; faithfulness is unnecessary.
Latent projection; Theorem 4	Projection misses marginalized observed common parents; POMIS equality relies on restoration → expanded-DAG projection, universal border dominance, and witness transfer.	POMIS equality holds at graph-only scope above. One witness preserves all allowed comparisons simultaneously. Equality of all interventional distribution classes is not asserted or needed.
bMVWA; Algorithm 2	Bootstrap weighting gives an unbiased minimum-variance estimate and its variance → minimize an estimated covariance objective.	The fixed-weight formula does not establish unbiasedness or true minimum variance for data-dependent weights. Plug-in estimates and Beta smoothing may be biased.
Beta matching; Algorithm 3	Use Beta matching and effective counts without full validity conditions → require $0 < m < 1$, $0 < v < m(1 - m)$; otherwise use direct-arm Beta moments.	No automatic posterior calibration, coverage, or inherited KL-UCB regret guarantee. Bernoulli rewards and finite arms are assumed.
Algorithms 2–3; estimation procedure	Underspecified bootstrap count, failed identification, support, initialization, and fallback → pass the count, omit failed/undefined estimators, and specify direct-arm initialization and fallback.	Required population support remains an assumption. Revised procedures did not generate the retained simulations; their numerical effect is unmeasured.
Throughout; empirical evaluation	Imprecise wording, references, and motivating-source descriptions → correct them; compute improvement percentages from rounded table entries.	Original table entries and plots are unchanged. Percentages describe displayed entries, not unrounded simulation outputs.

Experimental scope. Monte Carlo experiments were not rerun. Displayed identifying formulas were checked against direct SCM calculations on tested models. Original plots and table entries are retained; revised initialization, support checks, and fallback rules have not been evaluated in those experiments.

Citation. Cite the original publication above. Add a *note* identifying the *Unofficial AI-corrected version dated 2026-09-19* consulted and this URL: https://www.sanghacklee.me/ai-corrected/SCB2019_AI_corrected_2026-09-19.pdf. Keep the original publication year.

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